

Constructing the normalized trace on the completed CAR algebra

MathlibAnnex.CStarAlgebra.CAR.trace

def

Carries compatible bounded matrix traces through the algebraic limit, normed union and completion.

Statement

Let $A_n = M_{2^n}(\mathbb{C})$ with connecting maps $a \mapsto a \otimes I_2$, and let $\iota_n : A_n \rightarrow A$ embed the stages in the completed CAR algebra. The trace constructed here is the continuous complex-linear functional $\tau : A \rightarrow \mathbb{C}$ extending $\tau_n(a) = 2^{-n} \sum_i a_{ii}$. In particular, $\tau(\iota_n(a)) = \tau_n(a)$.

Definition

At a matrix stage of size $d = 2^n$, each diagonal entry satisfies $|a_{ii}| \leq \|a\|$. Consequently $|\tau_n(a)| \leq d^{-1} \sum_i |a_{ii}| \leq \|a\|$. Under $a \mapsto a \otimes I_2$, each diagonal entry is repeated twice and the normalizing denominator doubles, so the trace is unchanged. Iterating proves compatibility with every later-stage embedding.

The algebraic direct limit identifies two matrices when their images agree in a common later stage. Trace compatibility makes the value independent of the chosen representative. Transport this linear functional to the normed union. Every element there is represented at a finite stage; the stage embeddings preserve norm, so the same bound holds on the union.

For a Cauchy sequence x_k in that union, define the value at its limit by $\tau(\lim_k x_k) = \lim_k \tau(x_k)$. The bound makes the scalar sequence Cauchy and makes the limit independent of the representing sequence. This is the continuous extension used in the declaration.

Assumptions

The matrix norm is the C*-operator norm. The map $a \mapsto a \otimes I_2$ is a unital star homomorphism. Its entries with both added bit coordinates zero recover the entries of a , so it is injective; relabelling the coordinates does not affect injectivity. The C*-norm theorem for injective star homomorphisms then makes the stage map isometric. Thus the normed union has an unambiguous norm and its completion is A . These are proved features of the fixed construction, not new hypotheses about arbitrary directed systems.

Conclusion

The cited extension and stage-evaluation results identify one continuous functional on A . The contractivity, normalization, positivity, and trace-identity theorems show $|\tau(x)| \leq \|x\|$, $\tau(1) = 1$, $\tau(x^*x) \geq 0$, and $\tau(xy) = \tau(yx)$. Thus this extension is a normalized tracial state.

The declaration defines the continuous extension. Positivity, normalization and traciality are separate proved results, not conclusions established merely by giving the definition. Uniqueness among all normalized continuous traces is proved in the CAR trace-uniqueness theorem. No statement about extending this trace to the larger shell-generated target is part of this construction.

Main citations

- Exact defining declaration
- The completed CAR algebra
- Finite matrix stages
- Connecting embeddings by tensoring with the identity
- Injectivity of the connecting embedding
- Isometry of the connecting embedding
- Embeddings into later matrix stages
- Isometric stage maps into the normed union
- A common stage for representatives
- Normalized finite matrix trace
- The normalized matrix trace is contractive
- Compatibility with one amplification
- Compatibility with every later-stage embedding
- Trace on the algebraic direct limit
- Transporting the algebraic trace to the normed union
- Boundedness of the trace before completion
- The continuous trace on the normed union
- Agreement with finite-stage trace values
- Contractivity of the completed trace
- Normalization at the unit
- Positivity on adjoint squares
- The trace identity
- Injectivity of matrix amplification
- Uniqueness of the normalized continuous CAR trace

Lean source signature (exact)

```
noncomputable def trace : Limit →L[ℂ] ℂ :=
  preTrace.extend (UniformSpace.Completion.toComplL : PreCAR →L[ℂ] Limit)
```

In the source	Mathematical meaning
trace : Limit →L[ℂ] ℂ	The continuous complex-linear normalized trace $\tau : A \rightarrow \mathbb{C}$ on the completed CAR algebra A .
PreCAR; preTrace	The normed matrix-stage union A_{alg} and its bounded trace, whose value on a stage matrix $a \in M_{2^n}(\mathbb{C})$ is $2^{-n} \sum_i a_{ii}$.
UniformSpace.Completion.toComplL : PreCAR →L[ℂ] Limit	The canonical dense inclusion $A_{\text{alg}} \rightarrow A$.
preTrace.extend (...)	The complete RHS extends that same trace continuously to A , keeping $\tau(\iota_n(a)) = 2^{-n} \sum_i a_{ii}$. No competing trace or representation is an input; positivity, traciality and uniqueness are separate results.

The displayed definition includes its exact extension RHS. Its header-only extraction remains unchanged in the source-binding record.

Further source notes: The separately defined preTrace is the continuous trace on A_{alg} .

Earlier published Card and PDF

▼ Exact content identity

Declaration: MathlibAnnex.CStarAlgebra.CAR.trace
Accepted content SHA-256: 57264099d37f9dc44187a7311d8f6b02949c0531015973d46fd8ddaa0a6a632e
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Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73